

# On-Chain Parametric Cost Insurance for Underserved Latin-American Digital Workers

Record of Closed Iterations and Request for Methodological Guidance

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## §1. Motivation, problem, suggested solution, challenges

**Motivation.** We study Latin-American digital workers paid in Colombian pesos who carry recurring USD-denominated professional-input obligations (subscription AI services, API usage fees, cloud compute, software-as-a-service spend), a population whose macro-risk exposure is dominated by FX volatility: in emerging-market settings, exchange-rate movements are the primary transmission channel for inflation pass-through, interest-rate volatility, and capital-flow shocks (Calvo and Reinhart, 2002; Rey, 2015; Bruno and Shin, 2015).

**Problem.** When the COP weakens against the USD, the USD-denominated cost line translates into unbudgeted COP expenditure, and no Colombian retail-bank product currently hedges the resulting FX-pass-through cost shock at the hundred-to-low-thousands-USD monthly notional these workers carry.

**Suggested solution.** We identify the macro risks  $X$  that quantitatively produce these cost shocks, validate them empirically, and design parametric cost-insurance products that compensate for the FX-pass-through exposure (Carter et al., 2017; Mahul and Stutley, 2010; Clarke, 2016). The demand pattern we target — underserved-country households purchasing depreciation insurance in the form of a call option on the foreign-exchange rate — is documented in recent practitioner discussion ([Speaker — to be filled by user], 2025). The intermediation product compensates the worker in a unit she actually spends (COP, or a COP-pegged purchasing-parity stablecoin); the implementation venue is on-chain settlement infrastructure (described in the 37-page technical companion §4), not the economic primitive.

**Challenges.**

- Identification posture — predictive versus causal (see §4.2).
- Convex-payoff sufficiency under insurance-underwriting standard (see §4.2).

- Multi- $Y$  differential calibration; familywise versus false-discovery correction across legs (see §4.2).
- Anti-fishing protocol defensibility — pre-registration thresholds  $N_{\text{MIN}} = 75$ ,  $\text{POWER}_{\text{MIN}} = 0.80$ ,  $\text{MDES}_{\text{SD}} = 0.40$  (see §4.2).
- Cross-iteration familywise error and iteration-selection bias (see §4.2).

**Bridge to §2.** We have run five iterations to date, all closed or parked; the model specifications, results, and failure post-mortems follow in Section 2.

## §2. Five iterations

**Pair D — BPO offshoring through COP/USD lag (PASS).** Colombian young workers (14–28) employed in the broad CIU Rev. 4 services aggregate (Sections G–T), with  $Y_t$  the logit of the monthly young-worker share on DANE GEIH micro-data and  $X_{t-k}$  the lagged log Banrep TRM end-of-month spot.

$$Y_t = \alpha + \beta_6 X_{t-6} + \beta_9 X_{t-9} + \beta_{12} X_{t-12} + \varepsilon_t$$

where  $Y_t = \text{logit}(\text{share}_t^{\text{GT}})$  and  $X_{t-k} = \log(\text{COP/USD}_{t-k})$  for  $k \in \{6, 9, 12\}$  months; the composite is  $\hat{\beta}_{\text{composite}} = \hat{\beta}_6 + \hat{\beta}_9 + \hat{\beta}_{12}$ . Data: `notebooks/bpo_offshoring_fx_lag/` (governing spec sha 964c62cca0...ef659,  $N = 134$  monthly, 2015-01-31 through 2026-02-28). Result:  $\hat{\beta}_{\text{composite}} = +0.13670985$ ,  $\text{HAC}(L = 12)$  standard error 0.02465,  $t = +5.5456$ , one-sided  $p \approx 1.46 \times 10^{-8}$ ; robustness rows R1–R4 produced zero sign flips and the four-row consistency classifier returned AGREE. *Interpretation.* Lagged-FX positively predicts broad-services young-worker share; gate cleared. *Caveats inherited from independent review.* The estimate hedges a correlation, not a causal identification of the BPO transmission channel; the lag effect is concentrated at the six-month horizon ( $\hat{\beta}_6 \approx +0.109$ , about 80% of the composite) rather than spread across the 6–12-month window; the post-2014 window over-represents the oil-collapse, COVID, and 2022–2024 Fed-tightening regimes; the verdict is sensitive to a single off-spec design choice (an orchestrator-brief variant including a `marco2018_dummy` in the primary returns  $\hat{\beta}_{\text{composite}} = +0.0815$  at  $p \approx 0.080$ ); and the dev-AI Stage-1 R2 Section-M reframe (below) suggests the operative sub-aggregate is consultancy/scientific/administrative services rather than Section-J ICT-narrow tech offshoring.

**dev-AI Stage-1 — Section J ICT through COP/USD lag (FAIL).** Colombian young workers (14–28) restricted to CIU Rev. 4 Section J (Information and Communication, Divisions 58–63), a strict subset of the Pair D cut, with the same monthly cadence, `FEX_C_2018` expansion factors, and lag panel.

$$Y_t = \alpha + \beta_6 X_{t-6} + \beta_9 X_{t-9} + \beta_{12} X_{t-12} + \varepsilon_t$$

where  $Y_t = \text{logit}(\text{share}_t^{\text{J}})$  and  $X_{t-k} = \log(\text{COP/USD}_{t-k})$ ; pre-registered  $H_1 : \beta_{\text{composite}} > 0$ . Data: `notebooks/dev_ai_cost/` (governing spec sha 7c72292516...751f5a,  $N = 134$  monthly,

2015-01-31 through 2026-03-31). Result:  $\hat{\beta}_{\text{composite}} = -0.14613$ , HAC( $L = 12$ ) standard error 0.0847,  $t = -1.726$ , one-sided  $p \approx 0.958$  against  $H_1 : \beta > 0$ ; R-consistency classifier MIXED with three of four R-rows agreeing on the negative primary sign. *Interpretation.* Sign-flipped FAIL: positive-sign expectation rejected at the Section-J narrow cut, not an underpowered null. *Why it fails.* The Section-J substrate does not exhibit the lagged-FX-pass-through-to-employment pattern the BPO-offshoring narrative pre-supposed; an empalme-residual bias surfaced (the Phase-1 boundary diagnostic returned a  $+0.375$  logit- $Y$  jump at the 2020-12  $\rightarrow$  2021-01 Marco-2018 boundary against a  $\pm 0.335$  envelope), so the post-2021 level shift is not fully neutralized at the Section-J cut. The R2 Section-M sensitivity arm (CIU Sections 69–75) returned  $\hat{\beta}_{\text{composite}} = +0.45482801$ ,  $t = +4.73$ ,  $p = 1.13 \times 10^{-6}$ , but this is a not-pre-registered candidate-next-iteration object, not a graduated PASS, and the rejection at Section-J falsifies the per-user instrument design that the dev-AI-paying population would have hosted.

**FX-vol-CPI surprise — Colombian CPI through TRM realised volatility (FAIL).** Colombian corporates and remittance recipients with FX-volatility exposure on COP/USD around DANE CPI publication, weekly cadence on a Friday anchor, with  $Y_w$  the weekly realised volatility of the Banrep TRM daily log-return panel and the regressor the lagged signed AR(1) CPI surprise.

$$RV_w^{\text{COP/USD}} = \alpha + \beta \text{Surp}_{w-1}^{\text{CPI}} + \varepsilon_w$$

where  $RV_w$  is realised volatility in week  $w$  and  $\text{Surp}_{w-1}^{\text{CPI}}$  is the lagged signed monthly CPI surprise mapped to weekly cadence (carried across the release week, zero otherwise), with a small set of macro-surprise controls in the primary. Data: `notebooks/fx_vol_cpi_surprise/` (panel fingerprint `nb1_panel_fingerprint.json`,  $N = 947$  weekly, 2008-01-02 through 2026-02-23). Result:  $\hat{\beta}_{\text{CPI}} = -0.000685$ , HAC(4) standard error 0.001794, 90% HAC interval  $[-0.003636, +0.002266]$  contains zero; the gate criterion  $\hat{\beta} - 1.28 \cdot \widehat{\text{SE}} = -0.002981 < 0$  fires the FAIL. *Interpretation.* No detectable weekly-RV response to CPI surprises under the pre-committed primary. *Why it fails and sensitivity caveat.* The pre-registered T1 exogeneity test rejects at  $F = 15.12$ ,  $p \approx 1.3 \times 10^{-9}$ , so the iteration is a predictive regression rather than a structural-causal one and  $\hat{\beta}$  cannot be read as an impulse-response. Two pre-registered sensitivities returned positive significant 90% intervals at the one-sided  $\alpha = 0.05$  convention — A1 monthly cadence  $\hat{\beta} = +0.0152$  on  $[+0.0057, +0.0246]$ ; A4 release-day-excluded  $\hat{\beta} = +0.0033$  on  $[+0.0005, +0.0062]$  — but neither is the iteration's main finding: the pre-registered weekly primary is the result, and A1/A4 sit as candidate inputs to a possible future monthly-cadence-primary specification rather than as a rescue of the weekly primary.

**Phase-A.0 Remittance — Colombian remittance corridor through FX (EXIT\_NON\_REMITTAN** Colombian household remittance recipients indexed against the Colombian remittance corridor with FX pass-through, with  $Y_w$  inherited from the FX-vol weekly panel and  $X_{w-1}$  a candidate on-chain remittance-flow proxy constructed from cCOP and COPm aggregate user activity on Celo.

$$RV_w^{\text{COP/USD}} = \alpha + \beta \Delta \text{Remit}_{w-1}^{\text{Col}} + \varepsilon_w$$

where  $\Delta\text{Remit}_{w-1}^{\text{Col}}$  is the lagged Colombian household-remittance flow change as proxied on Celo on-chain activity, partitioned to attempt to isolate household remittance from third-party-DEX activity, treasury roundtripping, and bot arbitrage. Data: Phase-A.0 Rev 4.1 spec chain (kill-criterion-bearing); window 2024-09 through 2026-04, weekly cadence (panel construction did not reach final  $N$  before the EXIT verdict fired). Result: no empirical estimate produced — exited on kill criteria pre-estimation. *Interpretation.* Specification never estimated; killed pre-estimation by  $k1 + \text{partial-k2}$  on 2026-04-24. *Why it exits.* Task 11.F Axis-1 event research returned zero of thirty peak-day flows fingerprinting as remittance, with approximately 87% of surveyed flow decomposing into TRM arbitrage, treasury roundtripping, campaigns, and bot activity; the cCOP/COPM aggregate user activity on Celo is third-party-DEX volume, not Colombian household remittance, so no post-hoc filter could reconstruct an X-driver that was never measured upstream. The exit is load-bearing on the kill-criterion discipline: absent the pre-registered EXIT, the rev-4 plan's non-stop filter iteration would have run to a filter that appeared to “rescue” a signal that was never captured.

**P1 Bittensor SN18 — event-study apparatus (PARKED).** The proprietary-AI policy-shock-exposed population (LATAM developers paying USD-denominated AI APIs as the downstream candidate; the apparatus pre-dates target-population narrowing), with  $Y_t$  a Bittensor SN18 (Cortex.t) alpha event-study indicator and  $X_t$  a set of pre-pinned policy-event dates (Bittensor halving 2025-12-15; dTAO mainnet activation 2025-02-13; Cortex.t milestones).

$$\text{AR}_{i,\tau} = \alpha_i + \sum_{\tau=-L}^{+L} \gamma_{\tau} \mathbf{1}\{t_i + \tau \in \mathcal{E}\} + \varepsilon_{i,\tau}$$

where  $\text{AR}_{i,\tau}$  is the event-window abnormal alpha indicator at relative time  $\tau$  for event  $i$ ,  $\mathcal{E}$  is the pre-pinned policy-event set, and the apparatus is evaluated over a nine-cell verdict cube (verdict-eligibility crossed with robustness-consistency), Bonferroni-corrected at  $\alpha_{\text{primary}} = 0.0167$  over the conjunctive family P1+P2+P3 with  $N_{\min}^{\text{events}} = 8$  and an asymmetric placebo gate. Data: spec sha `f855e036d3...b47aeab`; apparatus completed, data ingest deferred. Result: none — apparatus parked for the record on 2026-04-27; no graduating empirical claim produced. *Interpretation.* Slow-lane apparatus parked; no graduating empirical claim. *Why parked.* Apparatus complete; no graduating empirical claim. The user pivoted to the simple-composite- $\beta$  fast-lane track (Pair D, dev-AI Stage-1) and parked P1 to preserve the intellectual capital of the spec work, with re-activation conditions documented in the parking memory.

### §3. Risk-target framing

The risk target is FX-volatility exposure for a specific population: Latin-American digital workers paid in Colombian pesos (COP cash-flows) but carrying recurring USD-denominated professional-input obligations. When the COP weakens against the USD, the USD-denominated cost line translates into an unbudgeted COP-expenditure increase. The intermediation product target is a contract that compensates for this FX-pass-through cost shock, denominated in a unit the worker actually spends (COP, or a COP-pegged purchasing-parity stablecoin). The Abrigo Lemma's

abstract  $Y_{\text{inequality}}(t) = R_a(t) - R_c(t)$  framing — present in the 37-page technical companion §3 — is the eventual analytical generalisation; the entry-point letter introduces the concrete target only.

## §4. Asks

### §4.1 Suggested specifications

We invite the supervisor to suggest the  $(Y, X)$  pair, lag structure, sample window, primary inference convention, and robustness arms for the next iteration. Specifically: which sectoral cut should follow the Pair D / dev-AI G–T / J split given the R2 Section M signal; whether to repeat the FX-vol-on-CPI-surprise primary at A1 monthly cadence as the next pre-registration; or whether to pivot to a non-Colombian LATAM panel.

### §4.2 Risks

The bullets below are risks the project has already identified; we ask the supervisor to add risks we have not flagged.

- Identification posture — predictive-regression versus structural-causal; T1 exogeneity rejected on FX-vol; HAC fixed- $L$  per iteration.
- Convex-payoff sufficiency under insurance-underwriting standard — mean- $\hat{\beta}$  is necessary but not sufficient; [Carter et al. \(2017\)](#), [Mahul and Stutley \(2010\)](#), and [Clarke \(2016\)](#) set the standards.
- Multi- $Y$  differential calibration — joint-leg correlation, FWER versus FDR, regime-switching.
- Anti-fishing protocol defensibility —  $N_{\text{MIN}} = 75$ ,  $\text{POWER}_{\text{MIN}} = 0.80$ ,  $\text{MDES}_{\text{SD}} = 0.40$  inherited from Phase-A.0 Rev-5.3.1, not re-derived.
- Cross-iteration FWER and iteration-selection bias — five closed iterations; meta-multiple-comparison concern; [Heckman \(1979\)](#) as candidate frame.
- Stambaugh small-sample bias on persistent-regressor predictive regressions — [Stambaugh \(1999\)](#), [Phillips \(2014\)](#); log-FX is near-unit-root at monthly frequency.
- Empalme residual bias — DANE Marco-2018 methodology break not fully neutralised; surfaced in dev-AI Stage-1 NB02 Trio 1 boundary anomaly.

### §4.3 Bibliography

The current bibliography is documented in the 37-page companion’s Appendix B and is reproduced as `refs.bib`. We invite the supervisor to recommend additional references he considers load-bearing for the framework’s empirical work, particularly on (a) parametric-index insurance pricing in low-counterparty-credit-standing populations, (b) sectoral employment-share predictive regressions on real-effective-exchange-rate panels, and (c) cross-iteration false-discovery-rate correction regimes for observational pre-registration.

## §4.4 Connections

We invite the supervisor to introduce the project to other researchers, practitioners, or institutions whose work might complement this iteration roster — including but not limited to parametric-insurance underwriters with LATAM exposure, FX-derivatives operators serving sub-corporate notionals, on-chain settlement engineers, and applied econometricians working on premature deindustrialization in Latin America. The framework’s empirical work is open to collaboration; the supervisor’s network is the most efficient channel for identifying complementary contributors.

## References

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